Discussion week 8

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Nov 18, 2021

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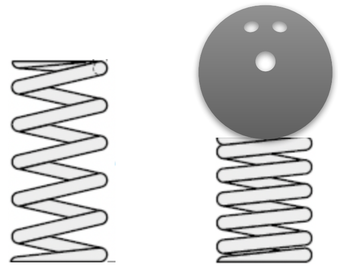
I.

1. An empty (assume massless) crate is on top of a stiff spring when a 5 kg bowling ball is placed slowly into the crate. The spring compresses 10 cm and the ball is at rest.



a. What is the spring's spring constant? $k = mg/\Delta x = 5 \times 9.8/0.1(N/m) = 490(N/m)$

b. Another bowling ball (m_2=10 kg) is slowly added to the crate. How much further does the crate drop?

$$\Delta x' = m_2 g/k = 98/490\ m = 0.2m$$

c. At this point (with two bowling balls in the crate), the crate is 1 meter above the ground. What is the entire potential energy of the two bowling balls? Include both elastic and gravitational potential energy.

$$\begin{aligned} E_p &= E_{sp} + E_g \\ &= \frac{1}{2}k(\Delta x + \Delta x')^2 + (m_1 + m_2)gh \\ &= 169.05J \end{aligned}$$

I.

Initial velocity is zero

d.One of the bowling balls (10kg) just barely rolls out of the crate. What happens to the remaining bowling ball in the crate?

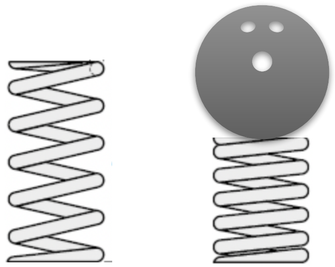
It moves vertically upward.

e.At what velocity does the ball which fell out of the crate hit the ground?

$$v = \sqrt{2gh} \approx 4.4m/s$$

f.How high above the ground will the second bowling ball (the one that stayed in the crate)?

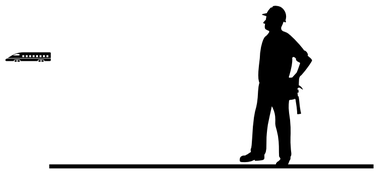
The energy of the system is conserved!



$$\frac{1}{2}k(\Delta x + \Delta x')^2 + m_1 gh = m_1 gh_{max}$$

2.

2.An Olympic biathlete (70 kg) stands on frictionless skis and shoots a rifle (mass = 5.0 kg) horizontally. The bullet has a mass of 0.02 kg and is fired at 1000.0 m/s.



a.What is the resulting velocity of the shooter and his rifle?

$$m_{bullet}v_{bullet} + (m_{rifle} + m_{shooter})v = 0 \qquad v = 0.27m/s$$

b.In which direction does he travel?

Opposite to the direction of the bullet

c.Which receives the larger impulse, the shooter+gun, or the bullet?

Same

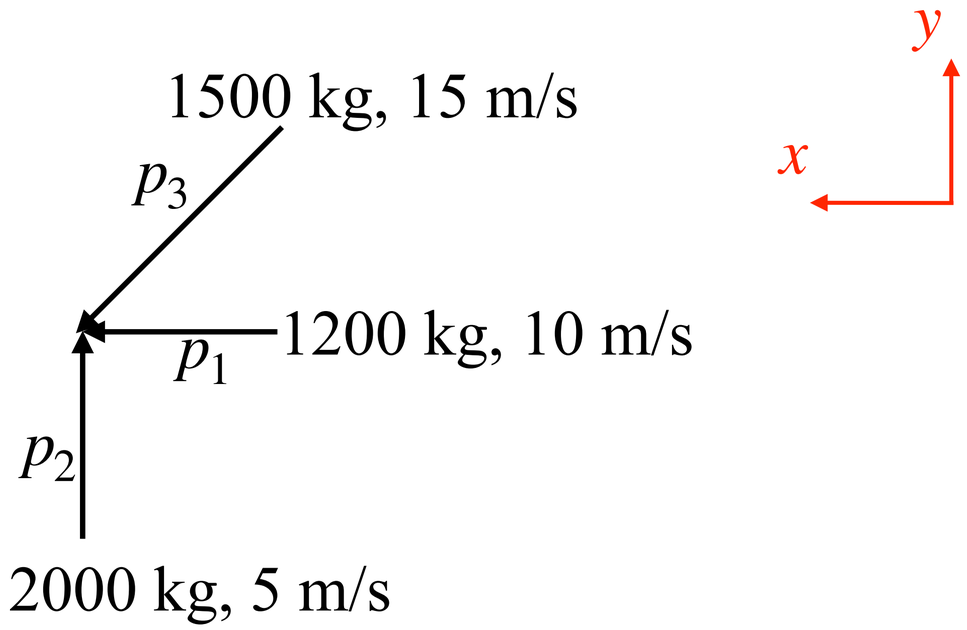
d.In reality, there is static friction and the biathlete will not move. Is momentum conserved? No

3.

3. A simultaneous three car collision occurs at an intersection. Car #1 (1200 kg) is driving West at 10 m/s. Car #2 (2000 kg) is driving North at 5 m/s. Car #3 (1500 kg) is driving South-West at 15 m/s. Assume the collision is "inelastic" such that the cars stick together after the collision.

Momentum is conserved

$$\vec{p}_{total} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3$$



3.

a. What is the magnitude of the velocity immediately after the wreck?

$$p_{tot,x} = p_3 \cos 45^\circ + p_1 = 27909.9(kg \cdot m/s)$$
$$p_{tot,y} = p_2 - p_3 \cos 45^\circ = -5909.9(kg \cdot m/s)$$

$$v = p_{tot}/m_{tot} = 28528.8/4700\ (m/s) \approx 6.0(m/s)$$

b. What is the direction of the three-car pile-up (give your answer in degrees W of N)?

$$\tan^{-1} \frac{p_y}{p_x} = -12.0^\circ$$

12 degrees south of west

c. The three cars, together, slide 10 meters after the collision, what is the coefficient of friction of the three car unit?

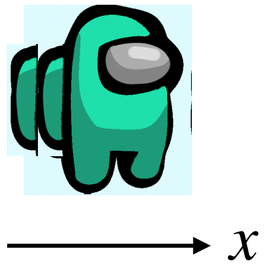
$$v^2 = 2\mu g \Delta s \rightarrow \mu = \frac{v^2}{2g\Delta s} \approx 0.18$$

4.

4. An astronaut (60 kg) is adrift far from her ship when the crew leaves her behind flying 10 m/s. All she has with her are two empty jet packs (20 kg each). She knows that, because of conservation of momentum, throwing the jet packs will propel her.

4.

a.If she threw the jet packs at the same time, how fast would she have to throw them (relative to her velocity) in order to catch the space ship? In what direction would she have to throw them?

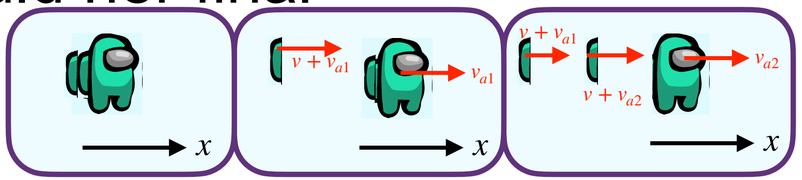


$$m_a v_a + 2m_j(v + v_a) = 0$$

$$v_a > 10m/s$$

$$v < -25m/s$$

b.If she threw them in sequence, one after another, at the same velocity (relative to her velocity) as in part (a), what would her final velocity be?



Step1 $(m_a + m_j)v_{a1} + m_j(v + v_{a1}) = 0$ $\quad v_{a1} = 5m/s$

Step2 $(m_a + m_j)v_{a1} = m_a v_{a2} + m_j(v + v_{a2})$ $\quad v_{a2} = 11.25m/s > 10m/s$

a.What is the cause of the discrepancy? Consider what the final velocities of the jet packs are when thrown sequentially.

$$v_{j1} = v + v_{a1} = -20m/s$$

$$v_{j2} = v + v_{a2} = -13.75m/s$$

As one can see, the total momentum of the 2 jet packs is larger when thrown sequentially. Therefore, the total momentum of the astronaut is larger in this case, which leads to a larger final velocity.